



Basic Statistics for Data Understanding

Turning Data Into Clear Summaries

Statistics is the practical language for summarizing data — not formula memorization, but clear answers to real questions.

What We'll Cover

01

Why Statistics Matters

Summarizing data beyond raw rows

03

Shape & Position

Distribution, skewness, percentiles, IQR

02

Center & Spread

Mean, median, mode, range, variance, standard deviation

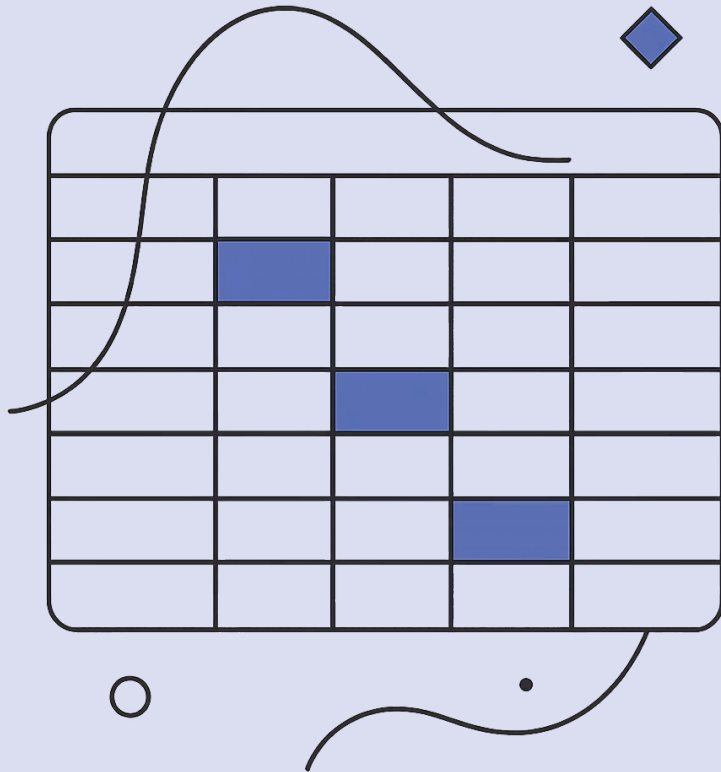
04

Relationships

Covariance, correlation, and plain-English observations

Rows Are Too Many to Read One by One

Even a small dataset becomes hard to understand row by row. Statistics gives us short answers to large tables.



Typical Values

What is common or average?

Spread

How different are values?

Relationships

Do columns move together?

Unusual Patterns

What stands out?

Start With Questions, Not Formulas

Each statistic is useful because it answers a practical question.

- What is the average monthly spend?
- What is the middle spend value?
- Which city appears most often?
- How different are customers from each other?
- Are visits and spending related?

The Four Statistical Lenses

Center, spread, position, relationship — the first four lenses for any dataset.



Center

What is typical? → Mean, Median, Mode



Spread

How different are values? → Range, Variance, Std Dev



Position

Where does a value sit? → Percentiles, Quartiles



Relationship

Do columns move together? → Covariance, Correlation

Our Working Example

We'll use this small customer spend dataset throughout the section. If you understand the tiny list, Pandas output becomes easier to read.

700

Customer 1
Monthly spend (₹)

850

Customer 2
Monthly spend (₹)

1200

Customer 3
Monthly spend (₹)

1450

Customer 4
Monthly spend (₹)

2200

Customer 5
Monthly spend (₹)

Mean

The Average — add all values, divide by count.

Step	Value
Sum	6,400
Count	5
Mean	1,280



The average monthly spend is 1,280 rupees. Mean is the balancing point of the values.

When Mean Is Useful — and When It Isn't

Mean Works Well When...

- Values are not extremely stretched
- You want one overall average
- The audience understands "average"

Mean Can Mislead When...

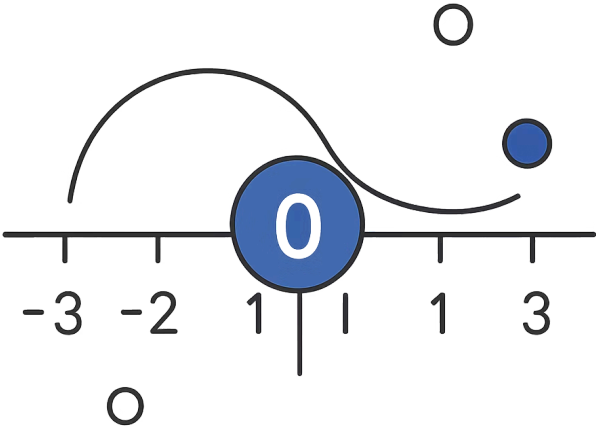
- One value is very high or very low
- Data is strongly skewed
- Extreme values pull the average

 Mean is useful, but it is sensitive to extreme values.

Median

The Middle Value — sort the values and take the middle.

Position	Value
1st	700
2nd	850
3rd (Median)	1,200
4th	1,450
5th	2,200



i Half the customers spend at or below 1,200; half spend at or above 1,200.

Mean vs. Median

When mean and median are far apart, ask whether unusual values are pulling the average.

Normal Values

700, 850, 1200, 1450, 2200

- Mean = 1,280
- Median = 1,200
- Close together — balanced data

One Extreme High Value

700, 850, 1200, 1450, 8,000

- Mean increases strongly
- Median stays more stable
- Gap signals an outlier

Choosing Mean or Median

Compare both before concluding. Mean tells average; median tells middle.

Situation	Better Summary	Reason
Values are balanced	Mean	Average represents the group well
Extreme highs or lows present	Median	Middle value is more stable
Reporting carefully	Both	Difference tells a useful story

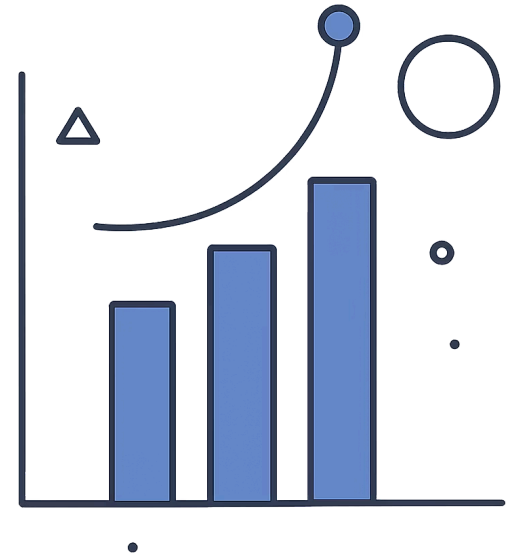
Mode

The Most Common Value — the value that appears most often.

Most useful for categorical columns:

- City
- Membership type
- Product category
- Signup month

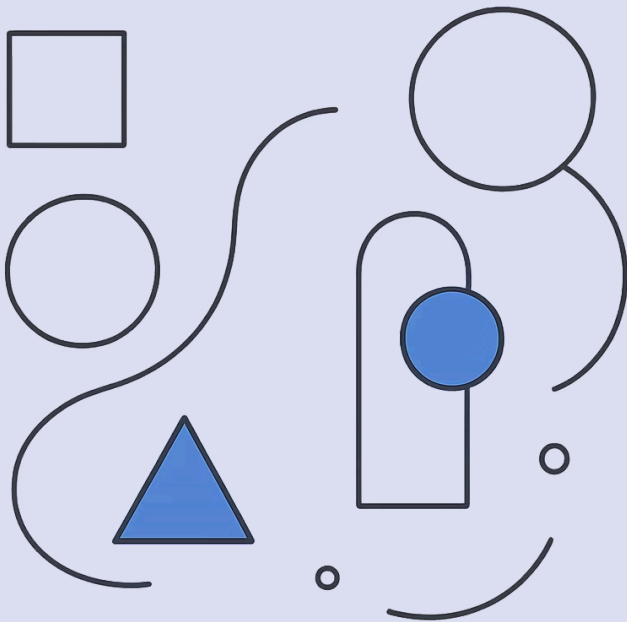
 Cities: Mumbai, Delhi, Mumbai, Chennai → Mode: Mumbai



Center Summary: Mean, Median, Mode

Statistic	Simple Meaning	Best For
Mean	Average	Numeric columns
Median	Middle value	Numeric columns with unusual values
Mode	Most common value	Categorical columns

 Do not use mean for categories. Do not use mode when you need the average of numbers.



Typical Value Is Not Enough

Two groups can share the same average but have very different spread.

Group A

900, 1000, 1100

Mean = 1,000 — tightly clustered

Group B

200, 1000, 1800

Mean = 1,000 — widely spread



Average tells center. Spread tells consistency.

Range

Maximum – Minimum. Easy to understand, but depends only on the two extreme values.

700

Minimum

2200

Maximum

1500

Range

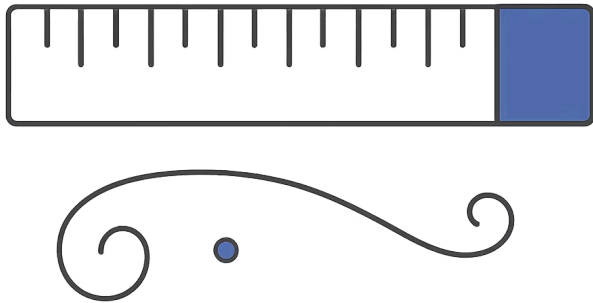
 Monthly spend values cover a range of 1,500 rupees.

Percentiles

A percentile tells where a value sits in sorted data.

Percentile	Name	Meaning
25th	Q1	25% of values at or below
50th	Median	Half of values at or below
75th	Q3	75% of values at or below

❏ Percentiles describe position after sorting.



Quartiles and IQR

The Interquartile Range ($IQR = Q3 - Q1$) measures the spread of the middle 50% of values.

Q1 — 25th Percentile Lower boundary of the middle half	Q2 — Median The midpoint of the data
Q3 — 75th Percentile Upper boundary of the middle half	$IQR = Q3 - Q1$ Less affected by extremes; useful for outlier checks

Worked Example: Customer Spend

Quartile	Value	Meaning
Sorted data	700, 850, 1200, 1450, 2200	Customer spend dataset in ascending order
Q1	850	25th percentile; lower boundary of the middle half
Q2 / Median	1200	50th percentile; middle value
Q3	1450	75th percentile; upper boundary of the middle half
IQR	600	$1450 - 850 = 600$

 IQR = 600 means the middle 50% of customers spend within a 600-unit range.

Variance

Variance measures spread using squared distances from the mean.

Worked Example: Customer Spend

Dataset: 700, 850, 1200, 1450, 2200

Mean = $(700 + 850 + 1200 + 1450 + 2200) / 5 = 1280$

Value	Difference from Mean	Squared Difference
700	-580	336400
850	-430	184900
1200	-80	6400
1450	170	28900
2200	920	846400

Variance = $(336400 + 184900 + 6400 + 28900 + 846400) / 5 = 280600$

Why Square the Differences?

Raw differences cancel out — squaring turns distance into positive spread.

The Problem

Values below the mean have negative differences.
Values above have positive differences. Adding them can cancel to zero.

The Solution

Squaring makes all differences positive, gives more weight to far values, and allows spread to be measured mathematically.

Standard Deviation

Standard deviation is the square root of variance — it returns spread to the original unit.

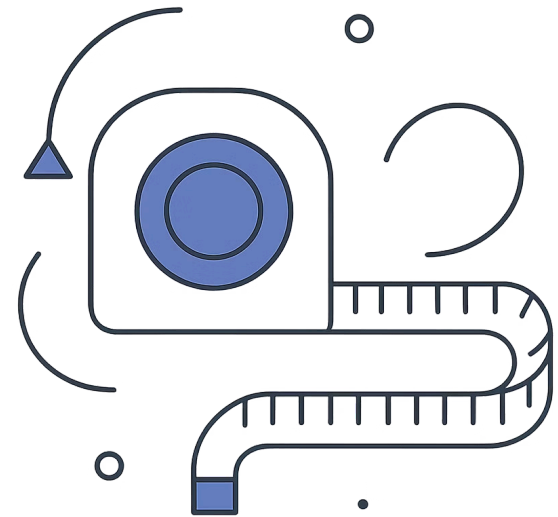
- Variance is useful for math
- Standard deviation is easier to explain
- If spend is in rupees, std dev is also in rupees

 Standard deviation = typical distance from the average.

Worked example: customer spend

Measure	Value
Variance	280,600
Standard Deviation	$\sqrt{280,600} \approx 530$

On average, customers spend about 530 rupees away from the mean of 1,280. So typical spend falls roughly between 750 and 1,810.



Variance vs. Standard Deviation

Measure	Meaning	Best Use
Variance	Average squared spread	Explaining the math idea
Standard Deviation	Typical spread in original units	Explaining to people

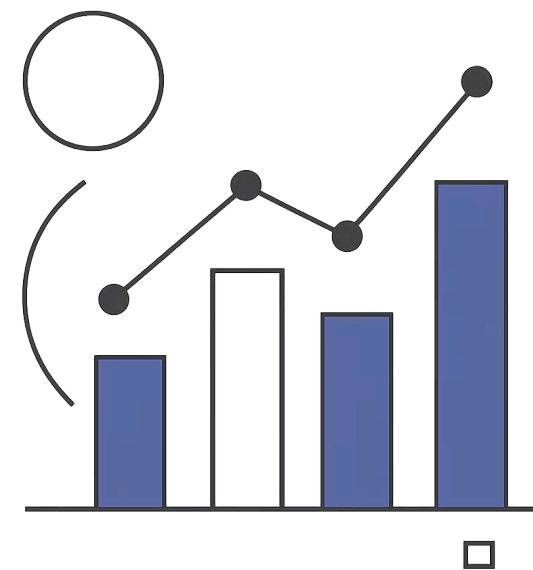
 Example: Customer spend varies by about 283 rupees from the average.

Distribution

Distribution describes the shape of the data — how values are spread across low, middle, and high ranges.

- Are most values low?
- Are most values in the middle?
- Is the shape balanced or stretched?

Common chart: Histogram



Worked Example: Customer Spend

Value	Position in Distribution	What It Tells Us
700	Low end	One customer at the bottom
850	Lower middle	Approaching the cluster
1200	Center	Middle of the data
1450	Upper middle	Top of the cluster
2200	High end	Stretches distribution right

Most customers cluster between 850 and 1,450. The value of 2,200 pulls the distribution rightward — a sign of positive skew.

Skewness

Skewness tells which side has the longer tail.

Skewness	Simple Meaning
Positive	Longer stretch toward high values
Negative	Longer stretch toward low values
Near zero	More balanced shape

 Example: Most customers spend 700–1,600, but one spends 8,000 — creating a positive (high-value) stretch.

Outliers and Statistics

This is why we compare multiple statistics instead of trusting one number.

Affected by Outliers

- Mean
- Range
- Standard deviation
- Charts
- Group comparisons

More Resistant

- Median
- IQR

✔ Median and IQR are robust choices when outliers are present.

Worked Example: What Happens When One Customer Spends 8,000?

Step 1 — Original dataset

700, 850, 1200, 1450, 2200

5 customers, no extreme values

Statistic	Value
Mean	1,280
Median	1,200
Range	1,500
Std Dev	~530

Step 2 — Add one outlier

700, 850, 1200, 1450, 2200, 8000

Statistic	Before	After	Change
Mean	1,280	2,400	↑ +1,120 (big jump)
Median	1,200	1,325	↑ +125 (barely moved)
Range	1,500	7,300	↑ +5,800 (exploded)
Std Dev	~530	~2,570	↑ +2,040 (very sensitive)

One unusual customer tripled the standard deviation and nearly doubled the mean — but the median barely moved. Always check both.

Do Two Columns Move Together?

Example question: *Do customers with more visits also spend more?*

Scatter Plot

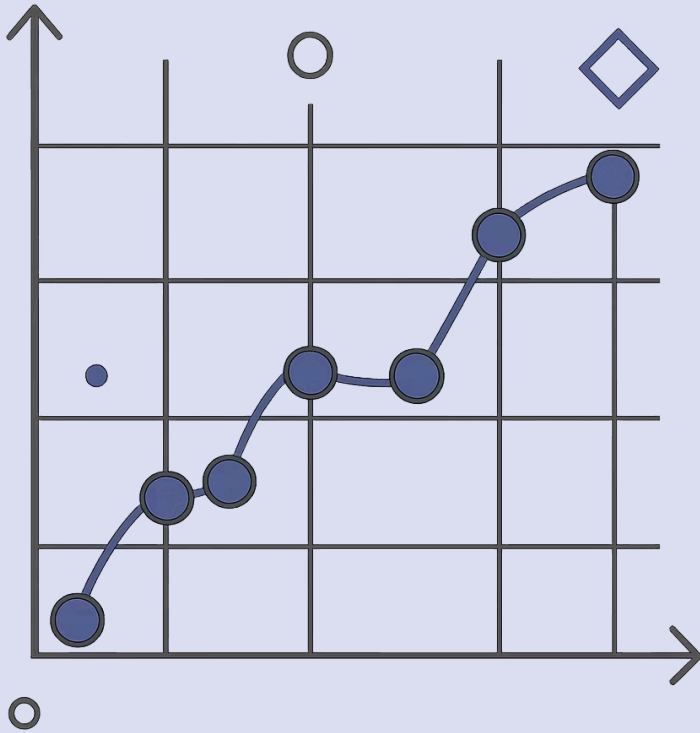
Visualize the relationship

Covariance

Direction of movement

Correlation

Direction and strength



Covariance

Covariance tells whether two numeric columns tend to move together. Note: the number depends on units, making direct comparison difficult.

Covariance	Meaning
Positive	Values tend to move in the same direction
Negative	One tends to rise while the other falls
Near zero	No clear same-direction or opposite-direction movement

Worked Example: Spend vs. Visits

Step 1 — Calculate means:

Mean spend = $(700+850+1200+1450+2200) / 5 = 1,280$

Mean visits = $(2+3+5+6+8) / 5 = 4.8$

Customer	Spend	Visits	(Spend – Mean)	(Visits – Mean)	Product
A	700	2	–580	–2.8	1,624
B	850	3	–430	–1.8	774
C	1200	5	–80	+0.2	–16
D	1450	6	+170	+1.2	204
E	2200	8	+920	+3.2	2,944

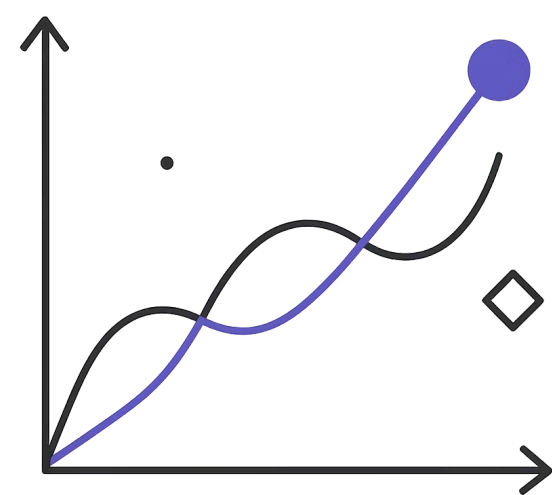
Step 3 — Covariance: $\text{sum of products} / n = (1624+774-16+204+2944) / 5 = 5530 / 5 = 1,106$

 Interpretation: Covariance = 1,106 (positive) → customers who visit more tend to spend more. But the number 1,106 is hard to compare across datasets because it depends on the units of spend and visits.

Correlation

Correlation standardizes the relationship between two numeric columns.
Range: -1 to +1.

Correlation	Meaning
Close to +1	Strong same-direction movement
Close to -1	Strong opposite-direction movement
Close to 0	Weak or no clear linear movement



❏ Correlation tells direction and strength.

Worked Example: Spend vs. Visits

Step 1 — Use the covariance result:
Covariance (spend, visits) = 1,106

Step 2 — Divide by the standard deviations:
Correlation = Covariance / (Std Dev Spend × Std Dev Visits)
= 1,106 / (530 × 2.14)
= 1,106 / 1,134
≈ 0.98

Measure	Value	What It Means
Covariance	1,106	Positive — same direction
Std Dev (Spend)	~530	Spread of spend values
Std Dev (Visits)	~2.14	Spread of visit values
Correlation	≈ 0.98	Very strong positive relationship

✔ Interpretation: A correlation of 0.98 means spend and visits move almost perfectly together in this dataset. Customers who visit more almost always spend more.

❏ Unlike covariance, 0.98 is easy to interpret — it's on a fixed -1 to +1 scale regardless of units.

Beginner Correlation Labels

Labels are guides, not universal laws. Always consider context and sample size.

Weak

Absolute correlation: 0.00 – 0.29

Moderate

Absolute correlation: 0.30 – 0.69

Strong

Absolute correlation: 0.70 – 1.00

 Example: Correlation = 0.559 → Moderate same-direction movement between visits and spend.

Correlation Is Not Cause

What Correlation Can Show

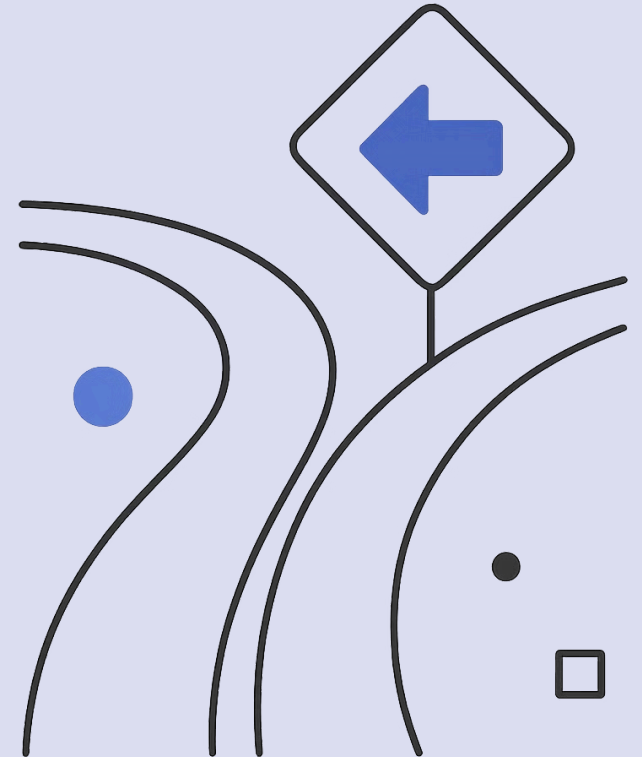
Visits and spending move together.

Other possibilities: high-interest customers both visit and spend more; membership level affects both; a promotion affected both columns; dataset may be too small to conclude cause.

What Correlation Cannot Prove

More visits caused higher spending.

⚠ Correlation is evidence of association, not proof of cause.



Grouped Statistics

Always look at group size — a high average from two rows is not strong evidence.

Membership	Count	Mean Spend	Median Spend
Gold	3	1,417	1,450
Silver	3	992	1,025
Bronze	2	863	863

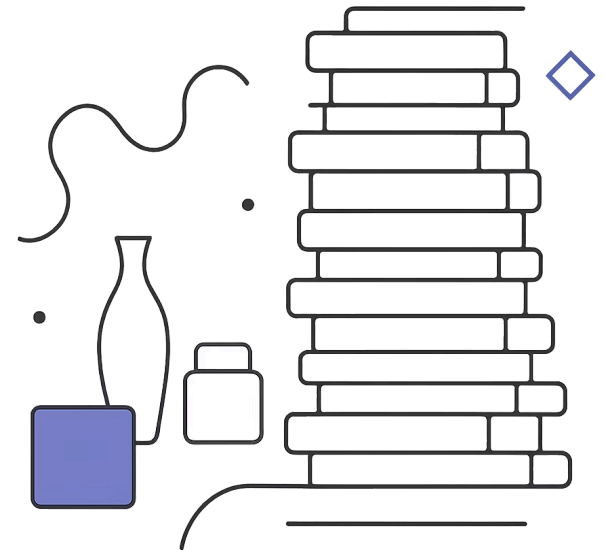
Small Dataset Caution

Small datasets are useful for learning the method, not for making big business claims.

- Patterns may not be stable
- One row can change the average
- Group comparisons can be fragile
- Correlation can be unreliable



Use small data to learn the method, not to draw strong conclusions.



describe() — Fast Numeric Summary

describe() is a dashboard for numeric columns, not the final analysis by itself.

```
customers[["monthly_spend",  
           "visits_per_month"]].describe()
```

Output Includes

- Count, Mean, Std Dev
- Minimum, Maximum
- 25th, 50th, 75th percentiles

Writing Observations From Numbers

A statistic becomes useful when you explain what it means.

Weak

The mean is 1100.

Better

The average monthly spend is 1,100 rupees in this cleaned customer dataset.

Stronger

The average monthly spend is 1,100 rupees, while the median is 1,025 rupees — check whether higher values are pulling the average upward.

Evidence-Based Observations

Pattern: Statistic + Value + Meaning



Median

The median monthly spend is 1,025 rupees, meaning half the customers spend at or below this value.



Standard Deviation

The standard deviation is 283 rupees, meaning spending varies noticeably around the average.



Correlation

The correlation is 0.559, suggesting moderate same-direction movement between visits and spend.

Common Beginner Mistakes to Avoid

Mean $\neq$ Median

Don't treat them as the same

Mean for Categories

Use mode instead

Ignoring Outliers

They affect most statistics

One Statistic Only

Always compare related summaries

Correlation = Cause

Association is not causation

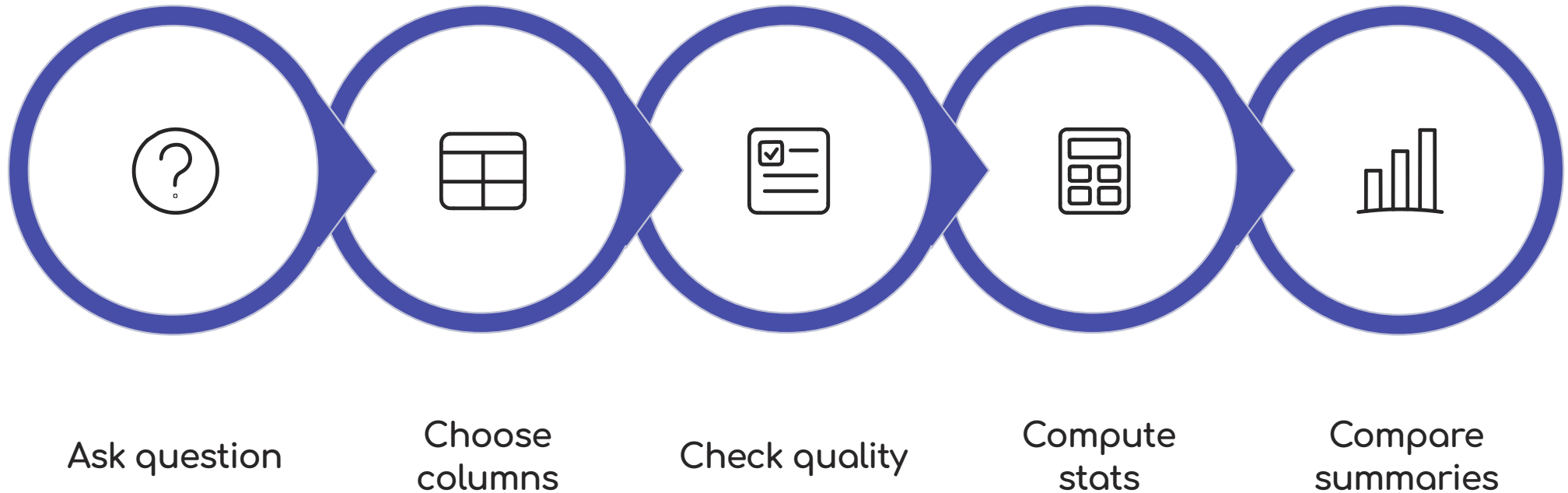
Ignoring Group Size

Small groups give fragile averages

📌 Statistics supports thinking. It does not replace thinking.

The Statistics Workflow

Statistics should be part of a workflow, not isolated calculations.



Notebook 04 — What You'll Practice

Notebook 04 turns these ideas into Pandas calculations and plain-English interpretations.



Center & Spread

Mean, median, mode, variance, standard deviation



Position

Range, percentiles, and IQR



Relationships

Covariance, correlation, distribution, skewness



Observations

Grouped summaries and evidence-based interpretations

Trainer Demo Flow

Teach the meaning first, then the Pandas method.

01

Start with a tiny spend list

Calculate mean manually

03

Explain spread with two groups

Show standard deviation as typical distance

02

Sort values and find median

Add one extreme value; compare mean and median again

04

Show correlation

Visits and spend; ask learners to write observations

Discussion Prompt

If the average customer spend is high, does that mean *most* customers spend high?

→ Not always

A few high spenders can raise the mean significantly

→ Check the median

Is it much lower than the mean?

→ Check distribution and range

Look for outliers pulling the average

☐ This is exactly why one statistic is not enough.



Section Summary

Statistics helps us move from numbers to understanding.

Center

Mean, median, and mode
describe typical values

Spread

Range, IQR, variance, and
standard deviation

Shape

Distribution describes shape;
skewness describes stretch

Relationship

Correlation shows direction and strength — not cause

Observations

Statistic + value + meaning = useful insight

Now We Calculate and Interpret

Next, we move into Notebook 04.

- Load the cleaned customer dataset
- Calculate common statistics
- Compare typical values and spread
- Read correlation carefully
- Write observations from output

✅ The goal is not to memorize formulas. The goal is to explain what the numbers say about the data.





Key Takeaway

Statistics is a language for summarizing data clearly and responsibly.

Ask a question

Every statistic answers something specific

Compare summaries

Never trust a single number alone

Explain the meaning

Statistic + value + plain-English interpretation